HW7.

Q1

reject H_0 if $\bar{X}_n > c$.

retain H_0 if $\bar{X}_n \leq c$.

Now we pin down c based on a given α.

under H_0, $\alpha = P_0(\bar{X}_n > c) = P_0\left(\frac{\bar{X}_n - \mu}{\sqrt{100/n}} > \frac{c-\mu}{\sqrt{100/n}}\right)$

$= 1 - \Phi\left(\frac{c-\mu}{\sqrt{100/n}}\right)$

$\frac{c-\mu}{\sqrt{100/n}} = \Phi^{-1}(1-\alpha)$

$c = \mu + \Phi^{-1}(1-\alpha) * \frac{10}{\sqrt{n}}$ → z_α. α-upper quantile. $z_{1-\alpha}$. $1-\alpha$ quantile

where $\mu = 110$ under H_0.

(a) if $\alpha = 5\%$, $n=16$, $c = 110 + z_{0.95} * \frac{10}{4}$, $z_{0.95} = 1.64$

≈ 116.1

$\bar{X}_n = 113.5 < c$

retain H_0. at the 5% size

(b) if $\alpha = 10\%$, $n=16$, $c = 110 + z_{0.9} * \frac{10}{4} \approx 113.2$

$z_{0.9} = 1.285$

$\bar{X}_n = 113.5 > c$

reject H_0 with 10% size

(c) p-val $= P_0(\bar{X}_n > \bar{x}_n) = P_0\left(\frac{\bar{X}_n - \mu}{\sqrt{100/n}} > \frac{\bar{x}_n - \mu}{\sqrt{100/n}}\right)$

$$= 1 - \Phi\left(\frac{113.5 - 110}{2.5}\right)$$

$$= 1 - \Phi(1.4)$$

Q2

(a) $H_0 : \mu = 3.4$

(b) $H_1 : \mu > 3.4$.

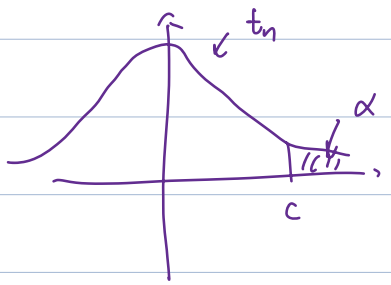
(c). under $N(\mu, \sigma^2)$ μ known, σ^2 unknown.

$$\frac{\sqrt{n}(\bar{X}_n - \mu)}{\sqrt{\frac{1}{n}\Sigma(X_i - \mu)^2}} \sim t_n.$$

test stat $T(X^n) = \dfrac{n(\bar{X}_n - \mu)}{\sqrt{\sum_{i=1}^{n}(X_i - \mu)^2}}$

(d). reject H_0, if $T(X^n) > c$.

for size α,



critical region is

$$\{ X^n : T(x^n) > t_n(95\%) \}$$

$t_n(95\%)$ is the 95% quantile of t_n.

Q3.

(a) under H_0, X_i's and Y_j's are iid $N(\mu, \sigma^2)$

$$\begin{bmatrix} \bar{X}_n - \mu \\ \bar{Y}_m - \mu \end{bmatrix} \sim N\left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} \sigma^2/n & 0 \\ 0 & \sigma^2/m \end{bmatrix} \right)$$

$$\bar{X}_n - \bar{Y}_m \sim N\left(0, \ \sigma^2\left(\frac{1}{n} + \frac{1}{m}\right)\right)$$

$$\hat{\sigma}^2 = \frac{1}{m+n-1}\left[\sum_{i=1}^{n}(X_i - \hat{\mu})^2 + \sum_{j=1}^{m}(Y_j - \hat{\mu})^2\right]$$

where $\hat{\mu} = \frac{1}{m+n}\left(\sum_{i=1}^{n} X_i + \sum_{j=1}^{m} Y_j\right)$.

$$\frac{\hat{\sigma}^2(m+n-1)}{\sigma^2} \sim \chi^2_{n+m-1}$$

$$\frac{(\bar{X}_n - \bar{Y}_m)\Big/\sqrt{\sigma^2\left(\frac{1}{n}+\frac{1}{m}\right)}}{\sqrt{\hat{\sigma}^2/\sigma^2}} \sim t_{m+n-1} \quad \text{under } H_0.$$

that is, $$T_n = \frac{\bar{X}_n - \bar{Y}_m}{\sqrt{\left(\frac{1}{n}+\frac{1}{m}\right)\hat{\sigma}^2}} \sim t_{m+n-1}$$

reject H_0, if $T_n > t_{m+n-1}(\alpha)$

α - upper quantile

$\alpha = 0.01$, $n = 16$, $m = 13$.

section 2.1 of Lecture 5 presents a test based on

$$T_n' = \frac{\bar{X}_n - \bar{Y}_m}{\sqrt{S_p^2\left(\frac{1}{n}+\frac{1}{m}\right)}} \sim t_{m+n-2}, \text{ under } H_0,$$

where $S_p^2 = \frac{n-1}{m+n-2}S_X^2 + \frac{m-1}{m+n-2}S_Y^2$.

Then. reject H_0, if $T'_n > t_{m+n-2}(\alpha)$.

(b) plug the numbers in T'_n,

$$T'_n = \frac{415.16 - 347.40}{\sqrt{\left(\frac{1}{6}+\frac{1}{13}\right)\frac{15}{27}\times 1356.75 + \frac{12}{27}\times 692.21}}$$

$$= 5.41$$

$$t_{m+n-2}(\alpha) = 2.479.$$

reject H_0 with size $\alpha = 0.01$.

(c) Welch's test stastistic (eq. 7.2-3)

under H_0,

$$W = \frac{\bar{X}_n - \bar{Y}_m}{\sqrt{\frac{1}{n}S_X^2 + \frac{1}{m}S_Y^2}} \sim t_{[r]}$$

where r is given in (7.2-1). $[r]$ is the floor of r.

$W \approx 5.677$

$r \approx 25.96$. $t_{25}(0.01) = 2.485$

$t_{26}(0.01) = 2.479$.

reject H_0, with size 1%